



APRIL 2024

REGIONAL MARITIME UNIVERSITY

MARINE ENGINEERING DEPARTMENT
BME 27B, BCE 27B, & BMT 27B.
END OF FIRST SEMESTER EXAMINATION
ENGINEERING DRAWING I

INSTRUCTIONS: ANSWER ALL QUESTIONS DATE: APRIL, 2024

TIME ALLOWED: 3 HOURS

QUESTION ONE

Given a straight-line $AB = 100$ mm, draw an equilateral triangle ABC . Maintain one of the apexes and rotate the triangle about it for six times. Name the resulting figure and dimension it properly. [15 marks]

QUESTION TWO

Figure 1 shows a dimensioned plan view of a gasket. Draw the given view showing all the tangent points clearly. [15 marks]

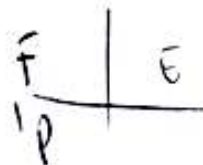
QUESTION THREE

The drawings in Figure 2 show a lathe tool post, a top slide, a tee bolt with hexagonal nut and washer and two set screws. The tee bolt fits the tee slots in the top slide and passes through the tool post. When assembled, these parts are secured in position by the washer and hexagonal nut. The set screws hold various tool bits in the tool post slot.

Draw full size, fully assembled, in the first angle projection the following views:

- A sectional front elevation looking in the direction of arrow A, and taken through the center line of the tee bolt.
- A complete plan projected from the front elevation. Hidden details are required in the plan view only.
- Add six main dimensions to the drawing.

[30 marks]



NOTE: ALL QUESTIONS SHOULD BE ANSWERED ON A1 SIZE PAPER

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$$\begin{array}{r} 1 \\ 38 \\ + 38 \\ \hline 76 \end{array}$$

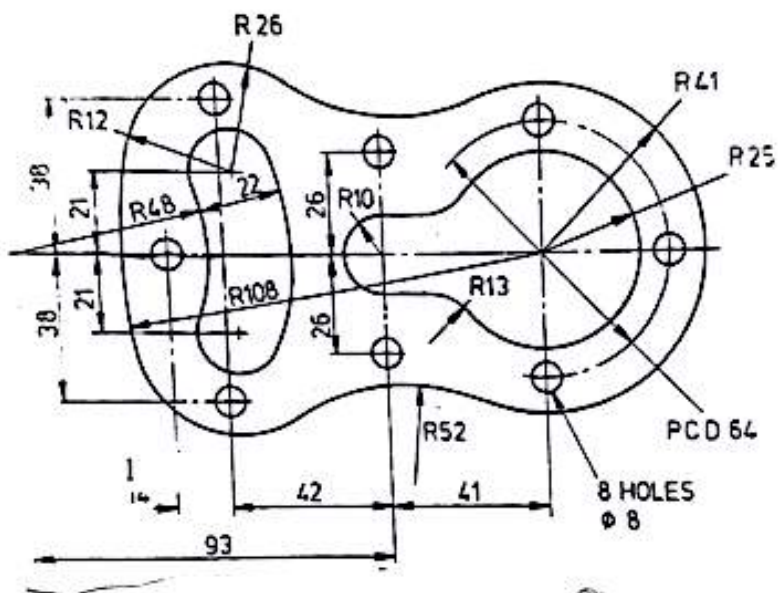


Figure 1

$$\begin{array}{r} 25 \\ 12 \\ \hline 37 \end{array}$$

$$\begin{array}{r} 108 \\ 52 \\ \hline 160 \end{array}$$

$$\begin{array}{r} 3 \\ 41 \\ \hline 44 \end{array}$$

7.6

13

$$\begin{array}{r} 108 \\ 26 \\ \hline 134 \end{array}$$

$$\begin{array}{r} 22 \\ 264 \\ \hline 286 \end{array}$$

$$\begin{array}{r} 76 \\ 193 \\ \hline 269 \end{array}$$

7

$$\begin{array}{r} 42 \\ 14 \\ \hline 56 \end{array}$$

$$\begin{array}{r} 2 \\ 12 \\ \hline 14 \end{array}$$

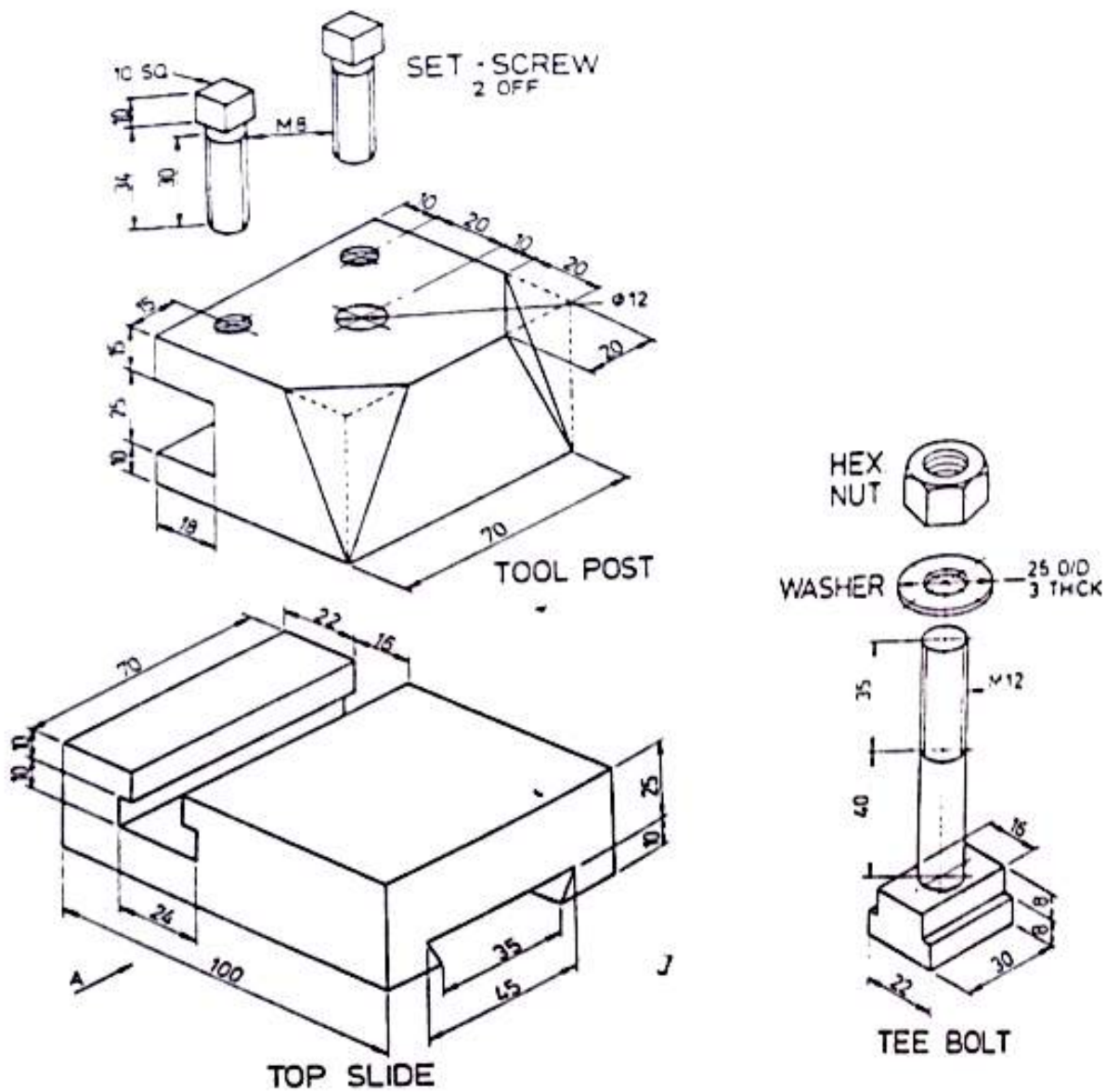


Figure 2

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